USN | | | | | | | | | | [22AS43]

FOURTH SEMESTER B.Tech EXAMINATION, MAY - 2024

AERODYNAMICS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART - A			**05x02=10**
1.	**a.**	Explain the following- Mach number	L2	CO1	**02**
	b.	Reynolds number	L2	CO1	**02**
	c.	Perturbation velocity	L2	CO1	**02**
	d.	Vortex tube	L2	CO1	**02**
	e.	Open system	L2	CO1	**02**
		PART – B			**06x05=30**
2.	**a.**	Define lift coefficient and drag coefficient. Obtain the expression for lift and drag force in terms of normal force 'N' and axial force 'A'. Write the expression for N and A.	L2	CO1	**02** **03**
		OR			
	b.	Explain solids, liquids and gases and their properties.	L2	CO1	**05**
3.	**a.**	Define coefficient of pressure (C_p). Obtain the expression for C_p using Bernoullis equations as $C_p=1-(V/V_\infty)^2$.	L2	CO1	**05**
		OR			
	b.	Obtain velocity potential and stream function for a source flow.	L2	CO1	**05**
4.	**a.**	Explain **i.** Open system **ii.** Closed system **iii.** Adiabatic process.	L2	CO3	**05**
		OR			
	b.	Define **i.** Reversible process **ii.** Isentropic process **iii.** Entropy	L2	CO3	**05**
5.	**a.**	Write the expression for induced velocity 'v' due to a finite vortex tube. Obtain the velocity at a point P, which is located at perpendicular distance of 5 m from a vortex tube of strength $\Gamma=20\ m^2/s$. **i.** for semi-infinite vortex tube **ii.** Infinite vortex tube.	L3	CO2	**05**
		OR			
	b.	Explain downwash velocity, induced angle of attack and induced drag.	L2	CO2	**05**

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6.	**a.**	Obtain the area velocity relationship for compressible flow.	L2	CO4	**05**
		OR			
	b.	Explain the flow through nozzle and diffuser for compressible flow as the Mach number changes from subsonic to supersonic.	L2	CO5	**05**
7.	**a.**	Write the expression for Velocity, V_r and V_θ for flow over circular cylinder. Show that $C_p=1-4*Sin^2(\theta)$ for inviscid flow over a cylinder.	L2	CO1	**05**
		OR			
	b.	Write the stream function for flow over uniform flow and doublet. Hence write the stream function for flow over cylinder.	L2	CO1	**05**
		PART - C		**06x10=60**	
8.	**a.**	Define source and vortex flow. Find out V_θ and C_p for flow over cylinder at $\theta=0°$ and $\theta=90°$.	L3	CO1	**04** **06**
		OR			
	b.	Define velocity potential and stream function. Obtain the governing equation for inviscid incompressible flow. State its properties.	L2	CO1	**04** **06**
9.	**a.**	Write the expression for C_l, C_d and C_m. An aircraft is flying at a speed of 50 m/s is having a wing area of 20 m^2. If the lift and drag coefficients are 0.6 and 0.03, find out the lift and drag force acting on the aircraft. Also calculate the resultant force.	L3	CO1	**05** **05**
		OR			
	b.	Define streamline, streak line and pathline. Obtain the location generator on cylinder non-lifting flow when the pressure equals free stream pressure.	L3	CO1	**06** **04**
10.	**a.**	Explain conformal transformation and Joukowski transformation. Explain the properties of conformal transformation.	L2	CO1	**04** **06**
		OR			
	b.	The Joukowski transformation is given by $\zeta=z+b^2/z$. Obtain the locations of singularity. State the necessary conditions for transforming a cylinder to **i.** flat plate **ii.** Symmetric airfoil	L2	CO1	**05** **05**
11.	**a.**	What is perturbation velocities and thin airfoil? Obtain the exact wall boundary condition over airfoil for invisicid flow in terms of free stream velocity and perturbation velocities.	L2	CO2	**04** **06**
		OR			

	b.	The boundary condition on an airfoil is given by $-(U\cos\alpha + u')\frac{dy}{dx} + U\sin\alpha + v' = 0$, where u′ and v′ are perturbation velocities and U is free stream velocity. Using proper assumptions, obtain the approximate boundary conditions. Explain the significance of camber problem and thickness problem.	L2	CO3	**05** **05**
12.	**a.**	Define **i.** Internal energy **ii.** Entropy **iii.** Boundary layer **iv.** Normal shock **v.** Oblique shock	L2	CO3	**10**
		OR			
	b.	Obtain the entropy change, s_2-s_1, for compressible flow in terms of C_p. Hence show that $p_0/p=(T_0/T)^{\gamma/\gamma-1}$ for isentropic flows.	L2	CO3	**06** **04**
13.	**a.**	Obtain the expression for displacement thickness and write the expression for momentum thickness. Explain boundary layer characteristics.	L2	CO6	**06** **04**
		OR			
	b.	Write salient properties of laminar flow and turbulent flow.	L2	CO6	**10**